

A PROJECT TITLE
ON
Customer Purchase Prediction Using Logistic Regression and Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF Computer Science and Engineering

Name of Title: Customer Purchase Prediction Using Logistic Regression and Random Forest

1. ABSTRACT:

Customer Purchase Prediction Using Logistic Regression and Random Forest is a machine learning project developed using Python to predict whether a customer is likely to make a purchase based on their available data. The system uses customer-related features such as age, gender, income, and previous purchasing behavior to identify purchasing patterns. Two supervised machine learning algorithms, Logistic Regression and Random Forest, are trained and evaluated to classify customers into purchase and non-purchase categories. The dataset is preprocessed using Python libraries such as Pandas and NumPy, while Scikit-learn is used for model training, prediction, and performance evaluation. The models can be compared using metrics such as accuracy, precision, recall, and F1-score. This project demonstrates how machine learning can be applied to customer behavior analysis and can help businesses make better decisions by identifying potential customers who are more likely to purchase a product or service.

2. INTRODUCTION:

Customer Purchase Prediction is a machine learning application used to predict whether a customer is likely to purchase a product or service based on information such as age, gender, income, and previous purchasing behavior. This project uses two supervised machine learning algorithms, Logistic Regression and Random Forest, to train models on customer data and predict purchase decisions. The models are evaluated using accuracy, precision, recall, and F1-score. The main objective is to compare both algorithms and identify the better-performing model while demonstrating how machine learning can help businesses understand customer behavior and make better decisions.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11, Linux, or macOS

Programming Language: Python 3.x

Development Environment: Jupyter Notebook or Visual Studio Code

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn

Algorithms: Logistic Regression and Random Forest

Hardware: Intel Core i3 or above, 4 GB RAM, and 1 GB free storage

Signature of the Guide

Course Instructor